

The Road Not Taken: College Board Geomarkets and Recruiting
Visits by Selective Colleges

ABSTRACT

Sociology of education investigates consumer-facing intermediaries (e.g., college rankings), but pays less attention to business-facing intermediaries that classify students on behalf of schools/colleges. The market devices literature offers useful concepts. Market devices are calculative tools incorporated into products or economic transactions. Performativity is the idea that actors utilize market devices in ways that enact behavior consistent with economic theories. College Board Geomarkets and the Market Segment model are market devices for higher education recruiting. Geomarkets carve metropolitan areas into smaller geographic units meant to define local recruiting markets. The Market Segment model predicts how student demand for a college varies by Geomarket as a function of social class. These market devices were the basis for the College Board Enrollment Planning Service, an early software-as-a-service product that helped colleges plan recruiting. Using data on recruiting visits to high schools made in 2017 by a sample of 42 colleges, we examine whether the relationship between recruiting visits and Geomarkets is consistent with theoretical expectations about performativity.

1 Introduction

Overarching RQ

- What is the relationship between College Board Geomarkets (X) and high school recruiting visit patterns?

Concrete RQs

- To what extent do college board Geomarkets explain which high schools receive recruiting visits from public research and selective private PSIs?
- To what extent are visit patterns consistent with recommendations from the Market Segment?
- If and to what extent does geomarket popularity [Z] mediate the relationship between school composition [free reduced lunch; share Black/Hispanic] and probability of a high school getting a visit [Y]?

2 Dependence and Patronage in the Enrollment Economy

The “enrollment economy” refers to reliance on student enrollment for financial stability. The concept was introduced by Clark (1956, 332), in his analysis of California adult education institutions: “The most important pressures on these schools in their day-to-day administration arise from the *enrollment economy* ...Financial support from the state is figured by the hours of attendance logged the previous year ...and a major slump in attendance constitutes a serious threat to organizational welfare.” (p. 332). At the opposite end of the prestige spectrum, Winston (1999, 17) describes dependence on students by selective private colleges: “the technology of producing much of what is sold in higher education is unusual in that colleges can buy important inputs to their production only from the customers who buy their products; that is, higher education uses a customer-input technology. Colleges depend on enrolled students for tuition revenue and on alumni for donation revenue. They depend on enrolled students for peer learning effects. Furthermore, the characteristics of enrolled students – for example, average SAT scores – substantially determine their national ranking, which then determines student demand for the subsequent cohort of prospects [CITE] and also affects research funding and donation revenue [CITE].

Stevens and Gebre-Medhin (2016) place the enrollment economy in historical context by tracing the sources of patronage across three “institutional configurations” – associational, national service, and market – that reflect major epochs in U.S. higher education.¹ The national service configuration, beginning after World War II, refers to massive federal and state investment in higher education as an engine of American political and technological development. Examples include federal research funding, state appropriations, and federal financial aid. The market configuration is associated with contemporary state disinvestment in public higher education, coupled with federal legislation encouraging research commercialization. Patronage – the adage, ‘he who pays the piper, calls the tune’ – is a central theme across all configurations. In the national service configuration, federal and state governments are the primary patrons. For example, in the analysis of California adult education by Clark (1956), state revenue is tied to student enrollment, not student ability to pay. In the market configuration, the primary patrons are households and private sector organizations. Here, the enrollment economy depends on identifying prospective students with high ability to pay. For colleges in the enrollment economy, organizational stability is the responsibility of enrollment management (EM). As a profession, EM integrates techniques from marketing and economics in order to “influence the characteristics and the size of enrolled student bodies” (Hossler and Bean 1990, xiv). As an administrative structure, the “office of enrollment management” typically controls and coordinates the functional activities of admissions, financial aid, and recruiting (Kraatz et al. 2010; Hossler and Bontrager 2014). The administrative structure of EM offers one means of categorizing scholarship about the supply-side of college access. In sociology, economics, and related interdisciplinary fields, the majority of scholarship focuses on admissions (e.g., Hirschman et al. 2016; Taylor et al. 2024; Posselt 2016; Killgore 2009) or on financial aid (e.g., Hurwitz 2012; Leeds and DesJardins 2015). A growing literature in sociology analyzes the earlier “recruiting” stages of acquiring “leads,” and soliciting “inquiries” and applications [CITE]. Several monographs analyze recruiting as part of broader studies on enrollment management (Stevens 2007), high school to college transitions (Holland 2019), elite reproduction (Khan 2011), or political economy (Cottom 2017). We situate the present manuscript within scholarship on recruiting that focuses on linkages

¹Early U.S. colleges were “associational” because they were initiated by religious or business associations and because colleges knit “themselves together in a wide variety of interorganizational associations” (Stevens and Gebre-Medhin 2016, 126) to solve common problems. Examples include voluntary accreditation, professional associations, and member associations like the Consortium on Financing Higher Education (COFHE).

between colleges and high schools (e.g., Holland 2019; Stevens 2007; Khan 2010).

The relationship between colleges and high schools is characterized by mutual – but often asymmetric – dependence. Colleges depend on a stable flow of high school students who collectively satisfy broad enrollment goals (Cheslock and Kroc 2012) – related to revenue, academic profile, and diversity – and the granular demands of internal constituents (Stevens 2007) – for example, the biology department needs majors and the band needs players. Although a college does not depend on any single high school to provide a substantial share of enrollment, the list of schools that enroll desirable students who are interested in attending is finite, thereby motivating “the recruiters’ emphasis on feeder high schools” (Zemsky and Oedel 1983, 11). High schools want to send their students to college. This goal is particularly important at expensive private schools and affluent public high schools, where placement into selective colleges functions as a key justification for both high tuition and high local housing prices. Dyadic relationships are seldom symmetric (Emerson 1962). A college values a high school to the extent that they want to enroll students from that high school. A private college that depends mostly on tuition revenue may place minimal value on a high school that primarily enrolls low-income students. A high school values a college to the extent that they want some of their students to enroll in that college. The distribution of student achievement suggests that high schools value a range of colleges. Colleges below this range may be viewed with disdain while colleges above this range may be viewed as unattainable.

Stevens (2007) provides an ethnography of enrollment management at a selective private liberal arts college. Being tuition-dependent and sensitive about its position in the rankings, The College focuses on enrollment goals related to academic profile and revenue generation. Stevens (2007) highlights the relational function of off-campus recruiting visits: “the College’s reputation and the quality of its applicant pool are dependent upon its connections with high schools nationwide” (p. 54). During the autumn “travel season,” admissions officers visit selected high schools across the country “to spread word of the institution and maintain relationships with guidance counselors” (p. 53-54). Ruffalo Noel Levitz (2020) reported that “calling high school counselors after visits” was an effective and widely-used outreach strategy for both private universities and public universities. Thus, relationship maintenance remains a high-touch process even in the digital age. The College tended to visit the same “feeder” schools year after year because recruiting depends on long-term

relationships. Visited schools tend to be affluent schools – in particular, private schools – that possess the resources to host a visit and enroll high-achieving students who can afford tuition.

Whereas scholars in the sociology of education have investigated recruiting as part of a broader analysis, AUTHORS (2021; 2022; 2024) have attempted to develop a literature that makes college recruiting behavior the object of empirical analysis. Salazar et al. (2021) argue that off-campus recruiting visits are a credible indicator of college enrollment priorities because recruiting visits cost resources and because visits to individual schools are largely invisible to macro stakeholders.² Salazar et al. (2021) analyzed recruiting visits made by public research universities. The authors found that 12 of the 15 universities made more visits to out-of-state schools than in-state schools. These out-of-state visits focused on affluent, predominantly white public high schools. Set against scholarship about nonresident enrollment growth as a response to declining state support (Jaquette and Curs 2015; Jaquette et al. 2016), these results suggest that many public research universities prioritize enrollment from wealthy out-of-state students.

Jaquette et al. (2024) conduct a social network analysis of recruiting visits to private high schools by a sample of public research universities and selective private research universities. They conceptualize visits as an indicator of a social relationship between a college and a high school, indicating that the college wants to enroll some students from the high school and that the high school wants to send some students to that college. Universities in the sample made a disproportionately large number of visits to private high schools relative to public high schools. Analyses revealed substantial overlap in the direct recruiting networks of public and private universities. Salazar (2022) conducted geospatial analyses of out-of-state recruiting visits to Los Angeles and Dallas by four public research universities. She found that universities engage in “recruitment redlining – the circuitous avoidance of predominantly Black and Latinx communities along recruiting visit paths” (p. 585).

We argue that scholarship on enrollment management has gone astray by focusing on college behavior as the source of racial and socioeconomic inequality in college access. Empirically, this literature focuses on the behavior of individual colleges. In turn, the literature assumes that enrollment management behavior is a function of individual college preferences, which are conditioned by

²Sociological theories of organizational behavior argue that organizations expend substantial resources on goals they care about while symbolically adopting – highly visible, ceremonial behavior with few resources expending – goals demanded by stakeholders in the broader institutional environment.

logics and forces in the organizational field (e.g., rankings, state funding). [DELETED BELOW SENTENCE ON KRIPPNER HERE] To our knowledge, scholarship on enrollment management has failed to consider third-party intermediaries that classify students on behalf of colleges. We present the case of Geomarkets and the Market Segment Model.

3 Geomarkets and the Market Segment Model

Geomarkets were conceived in the late 1970s, when colleges were concerned about declining demographics. Zemsky and Oedel (1983) [p. ix] write that demographer “Stephen Dresch (1975) had already frightened everyone who would listen with his projection of 40-percent declines in college enrollments by the close of the 1980s.” Elite colleges responded by leaning on the associational configuration (Stevens and Gebre-Medhin 2016) to solve problems common to the group. The project began when UPenn president, Martin Meyerson, tasked UPenn professor Robert Zemsky, to figure out, “‘Who thinks about Penn?’” and “‘What other institutions do they think about when they think about us?’” (Zemsky and Oedel 1983, x):

“In pursuit of these questions, we began working in 1978 with the Market Research Committee of the Consortium on Financing Higher Education (COFHE). The Consortium, a group of thirty selective private institutions, had in its brief history developed a remarkable tradition of sharing data...For nearly a year, the Consortium’s Market Research Committee provided focus to our efforts. Much of this project’s framework – the Market Segment Model and its derivation from admissions officers’ folklore was shaped and then reshaped in our discussions with members of the committee.”

Zemsky and Oedel (1983) conceived of the Market Segment Model as an effort to “capture and quantify” (p. 11) the knowledge of admissions officers: “We believe that the intuitions of admissions officers actually comprise a remarkably systematic body of knowledge about the college selection process...Our research is based upon listening carefully to what admissions officers have to say (pp. 9-10). The”initial [analytic] task” was to define geographic boundaries “in a manner consistent with admissions officers’ intuitive understanding of student pools” (Zemsky and Oedel 1983, 4). Quoting an admissions officer, Zemsky and Oedel (1983) [p. 11] write, “‘There are only three kinds

of college-bound students: those who want to live at home, those who want to live on campus but bring their laundry home, and those who want to go far enough from home that Mom and Dad can't visit without calling first.'” As such, Zemsky and Oedel (1983, 11) created “three types of boundaries – region, state, and community.” The initial regions in the project were New England, Middle States, and the South. Next, they “divided each state into as few as two and as many as thirty community-based enrollment markets or pools, for a total of 143 separate markets” (p. 11). These enrollment markets, later called Geomarkets, were intended to be consistent with the conception of a catchment market from the perspective of admissions counselors. Zemsky and Oedel (1983, 11–12) described the creation of Geomarket borders only briefly:

In many cases, the market boundaries match formal political and educational divisions, reflecting natural channels of communication. Each major metropolitan area is composed of several markets, usually corresponding to the inner city, a first ring of suburbs, and an outer ring of suburbs. In more sparsely populated areas, communities are sometimes combined in order to make the analysis meaningful.

Over time, the project goal morphed into predicting student demand for any college from any locality. However, “to gain a truly comprehensive view of the collegiate enrollment market, we needed a database that described most institutions and most students” (Zemsky and Oedel 1983, x). The group approached College Board. “Coincidentally, the Board was reviewing its own efforts to help colleges estimate their enrollment potential” (Zemsky and Oedel 1983, x). College Board provided score-sending data for all 1980 SAT test-takers. For a given student, each college that receives a score from the student can be defined as “local” (college located in the the same Geomarket as the student), “in-state” (same state but different Geomarket), “regional” (same region but different state), or “national” (different region). Zemsky and Oedel (1983) classified each SAT test-taker into one of four student *market segments*. A student is categorized in the “local” market segment if they submit more SAT scores to local institutions than they do to in-state, regional, or national institutions. An “in-state” student submits more SAT scores to in-state institutions than they do to local, regional, or national institutions, etc. The Market Segment Model “was nothing more than a set of simple rules for disaggregating high school seniors into similar groups” (Zemsky and Oedel 1983, 4). “The model worked because students, once so disaggregated, appeared to behave in

remarkably consistent ways.”

Zemsky and Oedel (1983, 3) investigates which student characteristics predict whether a student belongs to the local, in-state, regional, or national market segment. The analyses identify four variables – educational aspirations, parental education, scholastic aptitude, and family income – that predict score-sending behavior, both individually and in combination. The resulting thesis is that student demand for college is a function of social origin. These four variables “reflect the basic social patterns of the nation; it would have been surprising if these were not the four social variables that best explained the patterns of college choice” (Zemsky and Oedel 1983, 33). The analyses conclude (p. 42) that “our research has simply demonstrated what everyone has always known: communities with high levels of family income and parental education are also communities in which students have higher than average SATs and more far-reaching aspirations.”

However, Geomarkets differ in the relative abundance of students with particular socioeconomic characteristics. As college leaders sought new sources of patronage, the Market Segment Model yielded practical implications about which communities to target. Zemsky and Oedel (1983, 44) recommend that colleges target Geomarkets with desirable compositions of socioeconomic characteristics in order to reach students from desired student market segments:

On occasion, senior spokespersons for the profession worry that students outside the main [Geo]market areas remain forgotten and hence, unchallenged. Inevitably, the increasing competition for students, the expense of travel and mailings, and internal political constraints compel institutions to concentrate their efforts where they will do the most good. The result is a natural reinforcing of the basic socioeconomic patterns that gave shape in the first place to the structure of college choice.

Zemsky and Oedel (1983) developed two analytic outputs, (1) the Market Segment Profile and (2) the Institutional Profile. The Market Segment profile showed – separately by Geomarket – the number of students from each market segment (local, in-state, regional, national), their socioeconomic characteristics, and average test scores. The Institutional Profile showed – separately by Geomarket and institution (your own or a competitor) – how many students from each market segment sent scores to that institution.

In 1984, College Board transformed the Market Segment Model and associated outputs into the Enrollment Planning Service (EPS), an early software-as-service platform sold to colleges that was based on Zemsky and Oedel (1983). For any Geomarket, colleges could obtain the Market Segment Report and the Institutional Profile for themselves or a competitor. Based on background conversations with EM professionals, EPS software also provided information about the score-sending behavior of individual high schools within each Geomarket. Typical College Board (2005) marketing material describes EPS as, “the marketing software that pinpoints the schools and Geomarkets where your best prospects are most likely to be found. With the click of a mouse, EPS provides you with comprehensive reports on your markets, your position in those markets, and your competition.” Following Zemsky and Oedel (1983), the EPS platform encouraged users to, first, identify promising Geomarkets and, second, to identify desirable high schools for recruiting visits. To find new markets, EPS encouraged colleges to select peer colleges and identify Geomarkets where a large number of students sent scores to those colleges. Market research suggests that EPS was highly salient. Noel-Levitz (1998) reports that in 1995, 37% of 4-year publics and 49% of 4-year privates used EPS, while 41% of 4-year publics and 16% of 4-year privates used ACT’s market analysis service product.

4 Theory: Devices, Commensuration, and Classification

Market devices. We conceive of Geomarkets, the Market Segment Model, and the subsequent Enrollment Planning Service product as examples of “market devices” (Muniesa et al. 2007; MacKenzie and Millo 2003). Market devices are discreet calculative tools (e.g., algorithms, metrics, graphs, screens, rankings models, spreadsheets) that are incorporated into products or economic transactions. In this framework, neither market devices nor actors have agency in isolation from one another. “Agencements” are actions that become possible when actors utilize market devices. To make these concepts concrete, consider the case of US News College Rankings. The market device is the ranking tool itself – which consists of data inputs, an algorithm, and a ranked list published on a website. Agencements refer to the ways that actors utilize the rankings market device. For example, prospective students incorporate rankings in their college search process, employers use rankings to decide which colleges they will recruit from, and college administrators make resource allocation decisions designed to move up in the rankings (Espeland and Sauder 2016).

“Performativity” describes the consequences of market devices. Performativity is the idea that economic theories do not merely describe behavior, rather market devices are constructed and acted upon in ways that bring about behavior consistent with the theory (Muniesa et al. 2007; MacKenzie and Millo 2003).³ For example, MacKenzie and Millo (2003) provide a historical sociology of the Chicago Board Options Exchange, which opened in 1973. An option is a contract that gives the holder the right to buy or sell an asset (e.g., a stock) at a given price. Economists Fischer Black and Myron Scholes developed the Black-Scholes model – a calculative market device – to predict the price of options Black and Scholes (1973), eventually winning the Nobel Prize in Economics. Initially, the Black-Scholes algorithm did not accurately predict the price of options. Black-Scholes model predictions became more accurate over time because options traders began embedding the logic and components of the model into spreadsheets/software. That is, the model’s predictions became more accurate because the model produced the prices it predicted (MacKenzie and Millo 2003)! Bastedo and Bowman (2010) provide an example of performativity for the case of US News College Rankings. The reputational assessment of senior administrators was a substantial component of the rankings algorithm. Bastedo and Bowman (2010) show that administrators based their assessment scores on the prior overall rankings. Therefore, the rankings algorithm disciplines administrators to provide assessments that are consistent with the prior rankings.

Although canonical contributions to the market devices literature did not focus on structural inequality – race, gender, class, Hirschman and Garbes (2019) argue that market devices literature can offer novel insights about the mechanisms of structural inequality. Krippner (2024) offers an empirical illustration in her analysis of insurance products. Krippner (2024, 598) observes that prior feminist organizational scholarship considers gender discrimination an “organizational matter – a product of administrative routines such as hiring procedures, promotion protocols ...[but] this approach reinforced the notion that markets do not themselves contribute to discriminatory outcomes.” Drawing on documents from lawsuits filed by the National Association for Women, Krippner (2024) provides a historical document analysis that centers on the statistical models used to create life and car insurance products. She describes these models as “gendered market devices”

³Performativity is related to core concepts from the quantification literature, which has utilized college rankings as a case (Espeland and Sauder 2016). We highlight reactivity (people and organizations change behavior in response to salient quantitative measures), discipline (quantification disciplines actors to react in particular ways), and authority (quantification often weakens discretion of local decision-makers).

because gender is one of the input variables utilized to assign individuals to different “risk pools.”

The market devices literature offers great utility for the sociology of education subfield. The market devices literature examines how techniques, theories, culture, and agency interact in ways that make economic action possible (Hirschman and Garbes 2019; Fourcade 2007). Market devices are built upon other market devices and products are market devices that are composed of many market devices. This perspective legitimizes scholarly attention to individual components that are foundational to products and to markets (Fourcade 2011), facilitating scholarship that sacrifices breadth for depth. For example, rather than analyzing insurance products as a whole, Krippner (2024) focuses on the statistical models that assign people to different risk pools. Moreover, the market devices literature legitimizes scholarly attention to the technical details of how market devices are constructed because technical details have consequences (MacKenzie and Millo 2003; MacKenzie 2019). The flexibility of scope offered by the market devices literature is poignant for the sociology of education as third-party products increasingly perform core functions on behalf of schools and colleges. For scholarship that traces how a set of theories or techniques become embedded in one or more products, the market devices literature answers the essential question, ‘my study is a case of what?’ The market devices literature is situated within the economic sociology subfield. After some debate [CITE], it became part of the orthodoxy. Sociology of education draws heavily from organizational sociology (Trinidad 2024) but less so from economic sociology and, to our knowledge, has not substantively utilized the market devices literature as a foundation for empirical scholarship.

We conceive of College Board Geomarkets and the Market Segment Model as market devices that are based on previous market devices and become inputs into subsequent market devices. The SAT is a market device that enables (agencement) colleges to compare students from different schools. An additional agencement is that students can send their SAT scores to colleges they are interested in. The Market Segment Model is a theory [model? algorithm?] of student demand that categorizes students into different market segments based on SAT score-sending behavior. Geomarkets are a market device that defines geographic localities and enables colleges to compares these localities based on the number of students from each market segment. Geomarkets are then incorporated into the Enrollment Planning Service product – enabling colleges to decide which Geomarkets to

recruit from and which high schools to visit (Takamiya 2005) – and the Student Search Service product – enabling colleges to purchase the contact information of prospective students residing in desired Geomarkets. The high school visit patterns of colleges – the object of empirical analysis in this manuscript – speak to the performativity consequences of Geomarkets, the Market Segment Model, and EPS, whereby ...[state what we expect to happen]

Commensuration. The concept commensuration (Fourcade 2011; Espeland and Stevens 1998) helps us gain deeper insight about the agencement of market devices created by College Board. Commensuration is “the expression or measurement – of characteristics normally represented by different units – according to a common metric” (Espeland and Stevens 1998, 315). Commensuration is a process that requires work, akin to institutionalization. Once achieved, commensuration reduces the cognitive effort of decision-making. It is “a way to reduce and simplify disparate information into numbers that can easily be compared. This transformation allows people to quickly grasp, represent, and compare differences” (p. 316). For example, college rankings enable prospective students to compare colleges based on a single metric. Therefore, commensuration “can be understood as a system for discarding information and organizing what remains into new forms” (Espeland and Stevens 1998, 317).

The analysis of UK school “league tables” by McArthur and Reeves (2022) shows how commensuration can be a mechanism of social reproduction. In 1992, the UK government began publishing school league tables that ranked schools on student performance on national exams taken at age 16. League tables disciplined households – and in turn schools – to conceive of school quality in terms of test scores. League tables facilitated making evaluative comparisons between schools regardless of geographic proximity, but only in terms of the test score metric that was substantially a function of the class composition of schools. Following the adoption of league tables, localities with higher performing schools experienced growth in the share of managerial/professional households and a decline in the share of working-class households (McArthur and Reeves 2022). Therefore, league tables contributed to social reproduction because professional/managerial households were more aware of these consumer-facing metrics and had resources to respond by moving to more expensive neighborhoods, near higher performing schools.

Economics conceives of the SAT as a market device that facilitates commensuration. Hoxby (2009)

argues that “information costs” were a barrier to the emergence of a national higher education market that efficiently matches students to colleges: “In 1955, there was no early national college aptitude test. Students and colleges simply did not know where students stood in the national distribution of high school graduates’ achievement or aptitude” (p. 103). Colleges could not make an apples-to-apples comparison between students from different schools because the information on a high school transcript “is relative to a standard that a college will not understand unless it draws very often from the high school” (p. 103). Hoxby (2009) credits the SAT/ACT exams for the emergence of an efficient, national higher education market. Institutionalization of the SAT/ACT caused a “dramatic fall” in the cost of “colleges’ information about students” (Hoxby 2009, 102).

Geomarkets are a commensurating market device that builds on the SAT. Zemsky and Oedel (1983) created Geomarkets for the purpose of comparing different geographic recruiting territories. Consider the “MA10 – Milton, Lexington, Waltham” Geomarket versus “NJ10 – Bergen County.” How should we compare them, by population density, household income, their history of segregation? Zemsky and Oedel (1983) developed the Market Segment Model, which created four student segments: “local” students, who primarily sent SAT scores to colleges in their home Geomarket; in-state students; regional students; and national students, who sent SAT scores primarily to colleges outside their home region (e.g., New England). Whereas UK league tables allowed parents to compare geographically disparate schools based on test scores (McArthur and Reeves 2022), the Market Segment Model enables colleges to compare geographically proximate and geographically disparate Geomarkets based on the number of students from each market segment. Zemsky and Oedel (1983) states that selective colleges primarily draw students from the regional and national market segments. The key to finding new student markets, then, is identifying Geomarkets with large numbers of regional and national students. This information could be obtained from College Board EPS software.

Classification situations. Finally, scholarship on ordinal classification situations (Fourcade and Healy 2013, 2024, 2017; Fourcade 2016) helps conceptualize how Geomarkets facilitate social and racial stratification in college access. Fourcade (2016) defines three kinds of classification: *nominal* classifications, oriented to qualitative essence or “type” – such as race, gender, country of citizenship, or biological taxonomies; *cardinal* classifications, oriented to quantities and magnitude (e.g., SAT

score), where differences between quantities can be measured exactly but only for objects within the same nominal classification; and *ordinal* classifications, which are oriented to relative positions that can be hierarchically ranked, but the distance between positions are not measured quantitatively. Fourcade (2016) describes the growing salience of ordinal classifications. Objects once considered incommensurate because they are different nominal categories become ranked values of a single ordinal classification. For example Niche national private school rankings – utilized as a covariate in analyses below – commensurate a heterogeneous population of schools via a single ordinal ranking. The college-for-all movement is associated with the nominal binary of a college student being replaced by more inclusive, but hierarchical, ordinal classifications that ranges from the part-time adult learner in a community college to full-time Ivy League students.

Classification situations are ordinal interventions whereby organizations “sort and slot people into new types of categories with different economic rewards or punishments attached to them” (Fourcade and Healy 2013, 561). Classification situations intervene on the concept of class. Historically, sociology defines social class from the production side of the economy (e.g., worker vs owner; clerical, manager, professional). By contrast, Fourcade and Healy (2013) argue that processes of ordinalization “endogenous to the market” (p. 560) define class through effects on the *consumption* side of the economy. The canonical example is credit scores (Poon 2007), which enable third-party intermediaries on supply side to “slice consumers into behaviorally-defined risk groups, and price offerings to them accordingly. On the demand side, consumers find themselves ...fitting into these categories – which, by design, are not constructed from standard demographic classifications such as race” (Fourcade and Healy 2013, 560) but are highly correlated with social categories. The goal of classification situations is “good matches,” whereby an ordinal hierarchy of products match to an ordinal hierarchy of people. Businesses “seek good matches when they want to link a wealthy customer with a quality credit card, or a heavy drinker with a bad insurance plan” (Fourcade and Healy 2017, 17). Whereas economics tends to view such matches favorably – for example, Hoxby (2009, 106) writes the “efficient outcome allocates students to colleges based on students’ ability to benefit from the ...magnitude of the human capital investment that the college offers” – Sociology tends to highlight the predatory, extractive inclusion of formerly excluded consumers (e.g., for-profit colleges) (Cottom 2020).

Geomarkets and the Market Segment Model function together as ordinal market devices that match a socioeconomic hierarchy of places to a socioeconomic hierarchy of colleges.⁴ Zemsky and Oedel (1983) defined student market segments in ordinal terms – local, in-state, regional, and national – and found that student market segment was primarily a function of social class, particularly parental income and parental education. Zemsky and Oedel (1983) also conceived of colleges as an ordinal hierarchy. As in Fourcade and Healy (2017), good matches are the key to realizing enrollment goals. In turn, Zemsky and Oedel (1983) argue that good matches are defined by homophily, the idea that actors who share common characteristics are likely to form connections with one another (McPherson et al. 2001; Chun 2021). Zemsky and Oedel (1983) conduct a network analysis whereby two colleges are defined as competitors if a large number of students submit SAT scores to both colleges, Zemsky and Oedel (1983) “draw[ing] a fundamental conclusion about the structure of college choice: collegiate competition occurs principally between like institutions” (p. 46). Subsequent analyses investigate the tuition price and the socioeconomic composition of institutions in competition with one another. The takeaway is that like-colleges compete for like-students as defined by social class, with private selective colleges competing directly for students, charging the highest prices, and enrolling students with the highest socioeconomic status (p. 72):

Students describe themselves socially simply by telling us the colleges and universities in which they are interested. The layering of collegiate competition is primarily a socioeconomic layering. The hierarchical structure of collegiate competition largely reflects the stratified social and economic dimensions of the communities from which colleges draw their students. Competition among colleges, as admissions officers have told us for so long, is in fact, a matter of keeping company with one’s peers.

4.1 Research questions and expected results

Our overarching research question is, what is the relationship between college Board Geomarkets and high school recruiting visits of selective private colleges and public research universities? Because we lack the longitudinal data necessary for causal assertions, we investigate whether cross-sectional

⁴Whereas Fourcade and Healy (2024) analyze devices that classify individuals, the Market Segment Model harkens back to the earlier era of geodemography, in which market research classified places for merchants based on the attributes of people who lived there (Leyshon and Thrift 1999; McKelvey 2022)

2017 recruiting visit patterns are consistent with theoretical expectations about the performativity of Geomarkes and the Market Segment Model.

We disaggregate the overarching research question into three concrete questions. First, to what extent are visit patterns consistent with recommendations from the Market Segment Model? For selective private – and most public – institutions in our sample, the Market Segment Model recommends focusing on affluent Geomarkets located in regions where many students are likely to consider a college like yours (Zemsky and Oedel 1983).

In the contemporary “market configuration” of U.S. higher education (Stevens and Gebre-Medhin 2016), the enrollment economy for selective private colleges and the majority of public research universities depends on enrolling students willing to pay high tuition prices. The Market Segment Model is an algorithm for identifying the location of desirable patrons. Drawing from Fourcade (2016), the US postsecondary education system is highly ordinal and so are student market segments. According to Zemsky and Oedel (1983), selective private colleges primarily depend on students from the regional and national market segments. Further, regional and national students tend to come from affluent, college-educated households. Affluent households are not scattered randomly across the country, but rather clump together in particular geographic communities. Like UK “league tables” (McArthur and Reeves 2022), Geomarkets are a means of commensurating geographically disparate localities in terms of the number of students from each market segment. Imagine that the University of Arkansas has substantial regional but not national student demand and wants to enroll more out-of-state students. The Market Segment Model points to the affluent “TX 23 – Collin & Rockwall” (near Dallas) and the “TX 15 – Northwest Houston & Conroe School District” Geomarkets, which contain large numbers of regional and national students, but not the poorer “TX 19 – City of Dallas” or “TX 17 – City of Houston (East),” which contain relatively few regional and national students.

We answer RQ1 using simple descriptive statistics. We predict most colleges in our sample will adhere to a regional or national (but not an in-state) recruiting strategy, depending on whether they have regional or national enrollment demand. We also predict that colleges will disproportionately visit high schools located in affluent Geomarkets.

RQ2 asks, to what extent do College Board Geomarkets (X) explain which high schools receive recruiting visits (Y)? Drawing from the performativity concept, we argue that Geomarkets explain college recruiting visits to high schools – even after controlling for school- and neighborhood-level covariates thought to explain recruiting visits – because Geomarkets have become embedded in the architecture that produces recruiting visits. We highlight two mechanisms.

First, as described above, Geomarkets and the Market Segment model were the basis for the popular EPS software. Similar to the Black-Scholes model being written into spreadsheets used by options traders (MacKenzie and Millo 2003), Geomarkets and the logic of the Market Segment Model were embedded within EPS software, which encouraged colleges to identify promising Geomarkets and then select high schools to visit within these Geomarkets. Quantification often reduces the decision-making authority of local actors (Espeland and Stephens 2008). The Market Segment Model sought to replicate the aggregate knowledge of local admissions officers. On background, EM professionals told us that EPS software enabled colleges to plan travel without relying on admissions officers having strong local knowledge of their territories. EPS software often recommended visiting the same sets of affluent, high-achieving high schools that were receiving visits from other colleges. We were told that savvy, quantitatively-adept admissions offices used EPS to visit schools that EPS recommended ignoring because there would be less competition for the good students who attended these high schools.

Second, Geomarkets may explain recruiting visits because Geomarkets became an organizing principle for how admissions offices allocate admissions recruiters to territories. This process exemplifies performativity (MacKenzie and Millo 2003); Geomarkets were created to mirror the territories of admissions officers (Zemsky and Oedel 1983) and, over time, admissions offices structured recruiting territories around Geomarkets. On background, admissions officers and EM consultants told us that Geomarkets became part of the shorthand lexicon of the National Association of College Admissions Officers (NACAC). For example, admissions officers introduce themselves to one another by saying something like, “I recruit ‘PA 2,’” which refers to the “Chester County, PA Geomarket.” We find evidence for this claim scattered across the internet. For example, the LinkedIn profile of an admissions officer states, “Personally managed schools in Indiana geomarket’s 1, 2, 3, 4, 5, 6, 7, & 9” [CITE]. A report from the Dartmouth College Admissions Ambassador Program lists the number

of interviewers and applicants by Geomarket [CITE]. Although Geomarkets are utilized (agenced) by admissions professionals rather than prospective students, Admissions websites occasionally show concordance between Geomarkets and recruiting territories assigned to admissions officers. Examples include the Chicago-land recruiting territories of the University of Chicago [CITE], Lake Forest University, [CITE], and Saint Mary’s College [CITE]. Anecdotal evidence suggests that admissions offices allocate relatively more admissions officers to affluent Geomarkets. For example, the University of Chicago assigns one recruiter to the “IL 9 - North Shore” Geomarket, which had a 2020 population of about 161,000 and one recruiter to the “IL 12 - South and Southwest Suburbs” Geomarket, which had a 2020 population of about 1.2 million.

Additionally, we find evidence that higher education Customer Relationship Management (CRM) and Information System incorporate Geomarkets. Slate is the market-leading CRM for admissions and recruiting. Slate software allows colleges to assign admissions recruiters and prospective students to Geomarkets [CITE]. As another example, Oracle’s PeopleSoft Campus Solutions student information system product integrates Geomarkets (referred to as “EPS Market Codes”) and EPS. Oracle software documentation reads [CITE]:

EPS is a geographic and demographic data service offered annually by the College Board to Colleges and Universities. EPS provides information to subscribing institutions about competitors, feeder schools, and demographic strengths and weaknesses. EPS market codes are proprietary market codes owned by the College Board and are used to categorize external organizations and people into geographical areas, mostly in the United States. Some admissions offices use EPS market codes to focus their recruiting efforts in geographic areas in which they believe they will be the most successful.

RQ3 asks, to what extent does Geomarket popularity – measured as the number of visits per school – mediate the relationship between school composition and the probability of a receiving a visit? Previous research shows that having a high proportion of free-reduced lunch or Black/Hispanic students is negatively associated with receiving a visit (Salazar et al. 2021). Therefore, RQ3 investigates whether high- versus low-SES schools (and low versus high share of Black/Hispanic enrollment) benefit more from being located in a popular Geomarket. We hypothesize that high-SES and low Black/Hispanic enrollment schools benefit more from being located in a popular Geomarket.

Our rationale draws from Fourcade and Healy (2017), who highlight the role of “good matches” between ordinally classified products and ordinally classified customers. Fourcade (2016, 20–21) offers an anecdote:

One of the authors had a credit card transaction denied while trying to refill her car’s gas tank in a high-poverty Oakland neighborhood. The clerk indicated with resigned anger that the location of his gas station was the likely cause of the problem. Back in Berkeley a few minutes later, the card worked without a hitch.

Not coincidentally, Oakland is in Geomarket “CA 7 – City of Oakland” whereas Berkeley is in Geomarket “CA 8 – Alameda County excluding Oakland.” Fourcade and Healy (2017, 17) write that “‘good matches’ are perceived as natural because they fit well with how things already are.” An affluent school in an affluent Geomarket is a good match because it exemplifies the schools we expect to find. We predict that such schools benefit by receiving more visits than an affluent school in a less affluent Geomarket. By contrast, a poor or predominantly Black/Hispanic school in an affluent Geomarket does not match expectations and will not reap the same locational benefits.

Scholarly contribution. Sociology of education is concerned with socioeconomic and racial inequality. Empirical scholarship primarily locates the source of inequality in the behavior of households, schools, colleges, and policies (see reviews by Domina et al. 2017; Stevens et al. 2008). However, scholarship has an underdeveloped understanding of inequality caused by third-parties that mediate between consumers and sellers. Scholars have investigated intermediaries that “occupy the interface between consumers and producers” (Sharkey et al. 2023, 1), including scholarship on college rankings (Espeland and Sauder 2016) and school rankings (McArthur and Reeves 2022; Thompson 2024; Hasan and Kumar 2019). To our knowledge, intermediaries that classify consumers on behalf of sellers – akin to classification situations (Fourcade and Healy 2013) and including the field of market research – have rarely been the object of empirical research (but see Hamilton et al. 2024). Although consumer-facing intermediaries are highly salient to media outlets and researchers, intermediaries that classify consumers for businesses are more sophisticated and pervasive (McKelvey 2022; Muniesa et al. 2007; Cochoy 2007; Fourcade and Healy 2017) and may be highly salient to professionals responsible for identifying and recruiting customers.

This manuscript speaks to the performativity consequences of market devices by examining how College Board Geomarkets and the Market Segment Model mediate recruiting visits made by colleges to high schools. In doing so, we make three contributions to the sociology of education. First, to our knowledge, this is the first empirical analysis based on concepts from the market devices literature. We argue that the market devices literature offers great utility for future research that investigates the growing role of third-party vendors in schools and colleges. Second, whereas existing scholarship assumes that enrollment management behavior is a function of individual colleges (e.g., Salazar et al. 2021), we show how third-party market devices structure the recruiting behavior of individual colleges. Third, and most narrowly, we contribute to a critical sociology of the College Board. Existing research mirrors the blind-spot observed in scholarship on intermediaries by focusing on consumer-facing testing products [CITE]. This manuscript shows that College Board testing products are the basis for business-facing products that classify prospective students on behalf of colleges. Finally, analyses for RQ3 make a targeted contribution to the classification situations literature (Fourcade and Healy 2024) by assessing idea of “good matches” between producers and consumers.

5 Discussion

6 References

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